EXPERIMENT 3

3.1 Design and Comparative Analysis of Butterworth and Chebyshev Type-I IIR Band-Pass Filters Using MATLAB

MATLAB CODE:

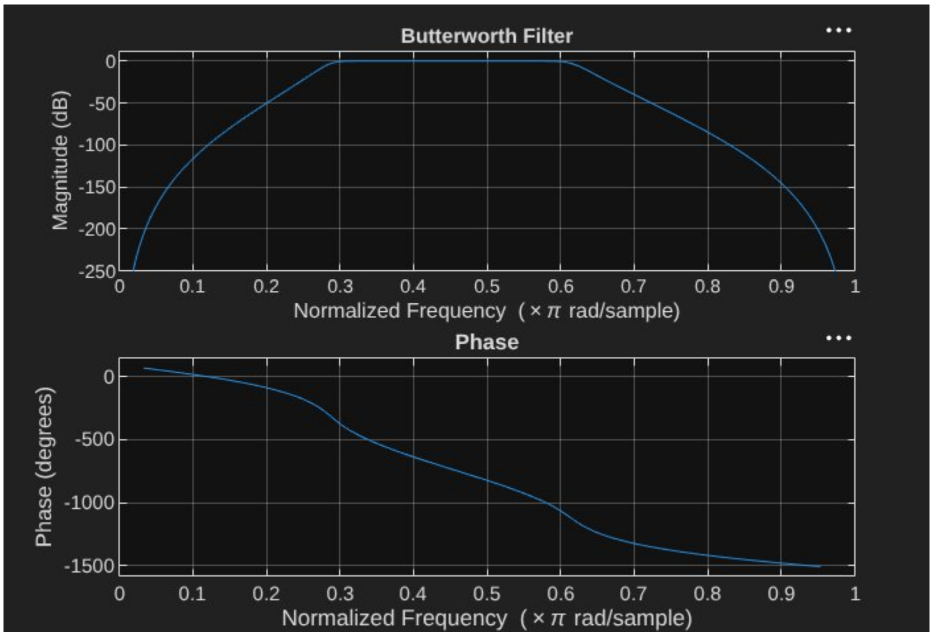
```
clc; clear; close all;
% Specifications
Fs=2000; Rp=1; Rs=40;
Wp=[300 600]/(Fs/2);
Ws=[200 700]/(Fs/2);

% Butterworth Filter
[n1,W1]=buttord(Wp,Ws,Rp,Rs);
[b1,a1]=butter(n1,W1,'bandpass');

% Chebyshev Type-I Filter
[n2,W2]=cheb1ord(Wp,Ws,Rp,Rs);
[b2,a2]=cheby1(n2,Rp,W2,'bandpass');

% Magnitude Response
figure;
freqz(b1,a1); title('Butterworth Filter');

figure;
freqz(b2,a2); title('Chebyshev Type-I Filter');
```

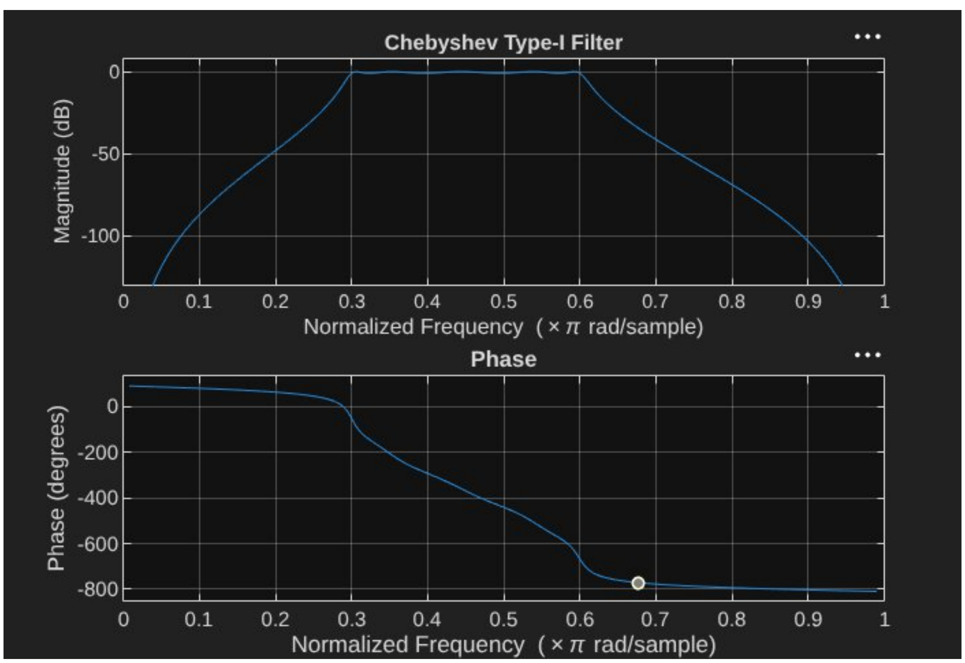
```
% Group Delay

figure;

grpdelay(b1,a1); hold on;

grpdelay(b2,a2);

legend('Butterworth','Chebyshev-I');


% Pole-Zero Plot

figure;

subplot(121),zplane(b1,a1),title('Butterworth')

subplot(122),zplane(b2,a2),title('Chebyshev-I')


% Impulse Response

figure;

subplot(211),impz(b1,a1),title('Butterworth Impulse')

subplot(212),impz(b2,a2),title('Chebyshev Impulse')


% Pole Magnitude

figure;

subplot(121),stem(abs(roots(a1)),'filled')

title('Butterworth Pole Magnitude')


subplot(122),stem(abs(roots(a2)),'filled')

title('Chebyshev Pole Magnitude')


disp(['Butterworth Order = ',num2str(n1)])

disp(['Chebyshev Order = ',num2str(n2)])
```

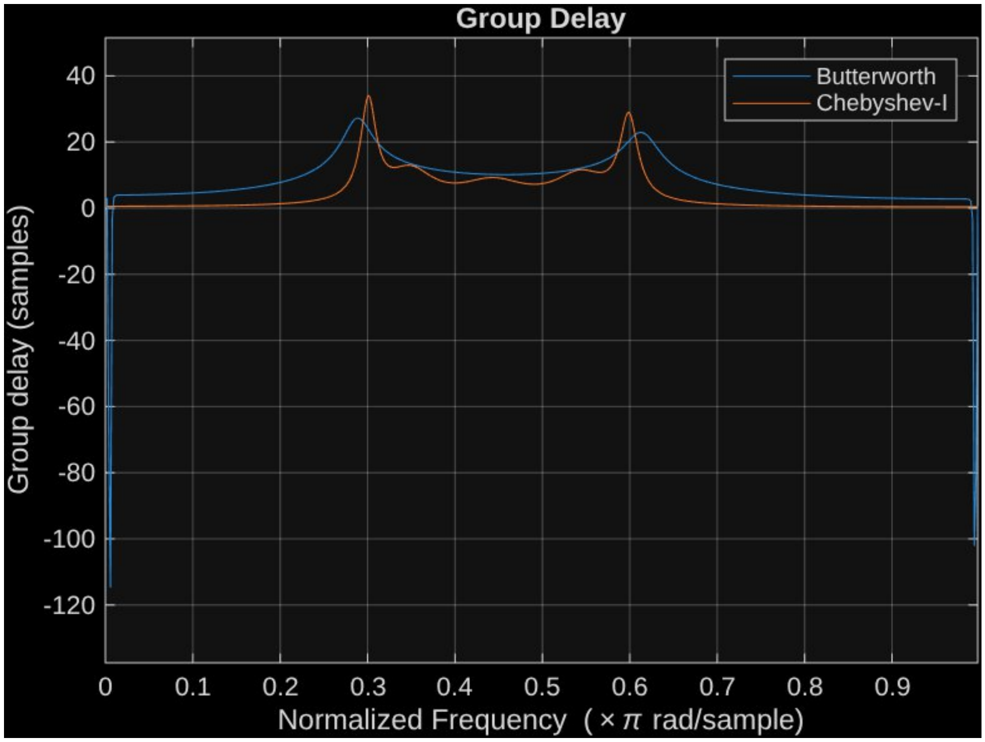
Command Window Output:

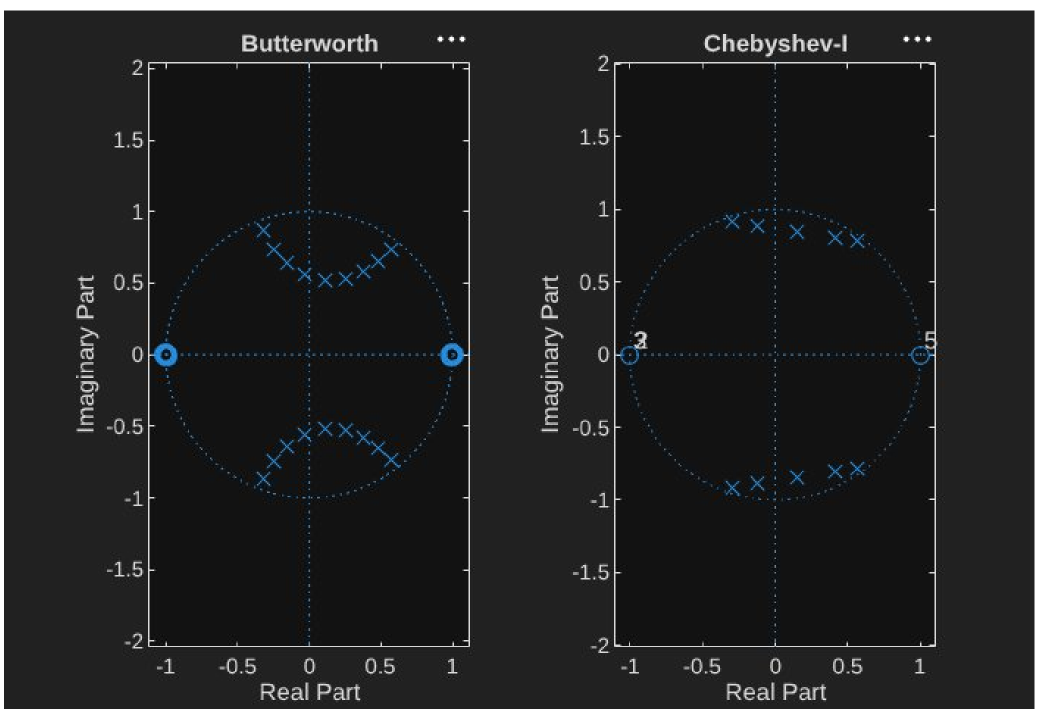
Butterworth Order = 9
Chebyshev Order = 5

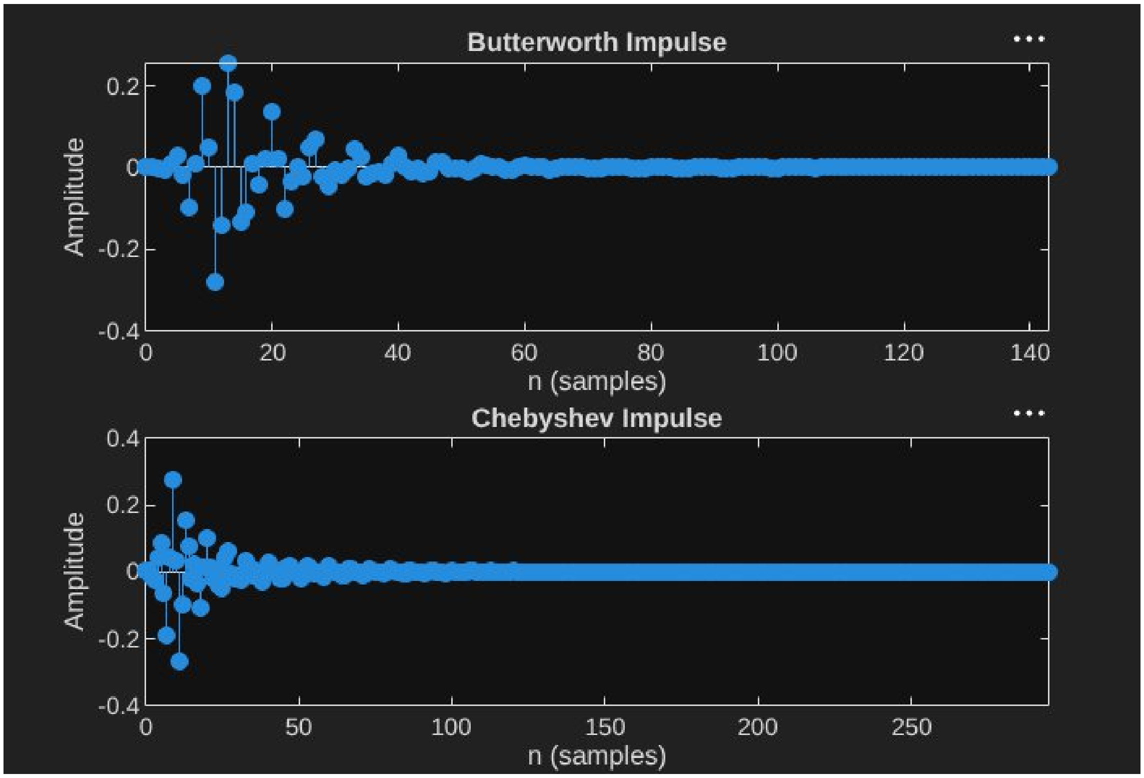
PLOTS:

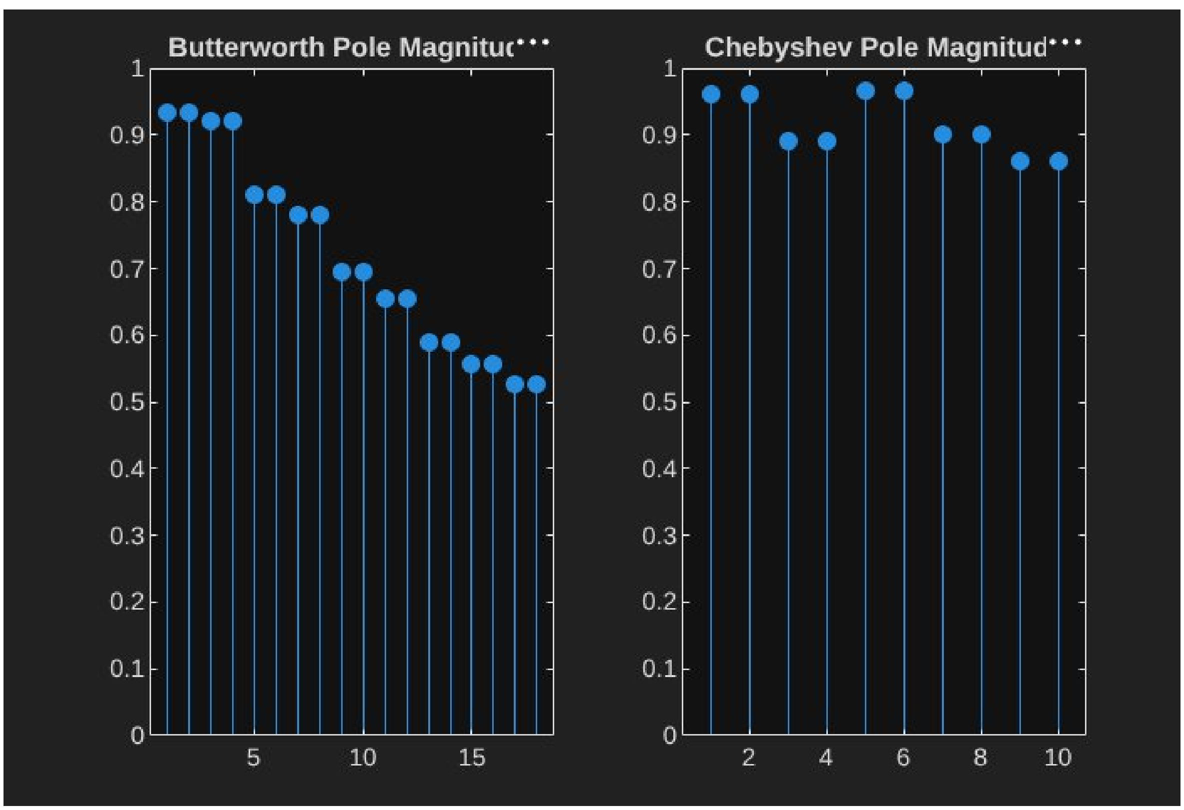





Group Delay
Butterworth
Chebyshev-I
Group delay (samples)
40
20
0
-20
-40
-60
-80
-100
-120
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
Normalized Frequency (× π rad/sample)


Butterworth
Chebyshev-I
Imaginary Part
Real Part
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
-1
-0.5
0
0.5
1
3
5


Butterworth Impulse
Amplitude
n (samples)
Chebyshev Impulse
Amplitude
n (samples)


Butterworth Pole Magnitu
Chebyshev Pole Magnitud